apply.) Here is the essential qualitative difference arising from the change in dimension.1 We recall from Chapter 11 that a pivotal realization is that strings, as they move through space, can lasso a two-dimensional sphere. That is, their two-dimensional worldsheet can fully surround a two-dimensional sphere, as in Figure 11.6. This proves to be just enough protection to keep a collapsing, pinching two-dimensional sphere from causing physical catastrophes. But now we are looking at the other kind of sphere inside a Calabi-Yau space, and it has too many dimensions for it to be surrounded by a moving string. If you have trouble seeing this, it is perfectly okay to think of the analogy obtained by lowering all dimensions by one. You can picture three-dimensional spheres as if they are two-dimensional surfaces of ordinary beach balls, so long as you also picture one-dimensional strings as if they are zero-dimensional point particles. Then, in analogy with the fact that a zero-dimensional point-particle cannot lasso anything, let alone a two-dimensional sphere, a one-dimensional string cannot lasso a three-dimensional sphere.

Such reasoning led string theorists to speculate that if a three-dimensional sphere inside a Calabi-Yau space were to collapse, something that the approximate equations showed to be a perfectly possible if not commonplace evolution in string theory, it might yield a cataclysmic result. In fact, the approximate equations of string theory developed prior to the mid-1990s seemed to indicate that the workings of the universe would grind to a halt if such a collapse were to occur; they indicated that certain of the infinities tamed by string theory would be unleashed by such a pinching of the spatial fabric. For a number of years, string theorists had to live with this disturbing, albeit inconclusive, state of understanding. But in 1995, Andrew Strominger showed that these doomsaying speculations were wrong.

Strominger, following earlier groundbreaking work of Witten and Seiberg, made use of the realization that string theory, when analyzed with the newfound precision of the second superstring revolution, is not just a theory of one-dimensional strings. He reasoned as follows. A one-dimensional string—a one-brane in the newer language of the field—can completely surround a one-dimensional piece of space, like a circle, as we illustrate in Figure 13.1. (Notice that this is different from Figure 11.6, in which a one-dimensional string, as it moves